
Phase 1 PDAC IND: AI Generation Draft 1.0

10.5281/zenodo.xxxxxxxx | ORCID 0009-0007-5457-8667
CEO Kevin Kawchak, ChemicalQDevice | kevink@chemicalqdevice.com

IND version: v1.0 **Repository version:** v4.3.0 **Date:** July 1, 2026 **San Diego**
Repository: kevinkawchak/physical-ai-oncology-trials/tree/main/trial-ind

Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference. Disclaimer: this work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 4.8. The DOI placeholder 10.5281/zenodo.xxxxxxxx is filled at deposit.

Cover Letter

ChemicalQDevice
San Diego, California
July 1, 2026

Food and Drug Administration
Center for Drug Evaluation and Research
Division of Oncology 1

Re: Initial Investigational New Drug Application, Serial 0000 - Daraxonrasib (RMC-6236), Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC

[DRAFTING INSTRUCTION] Draft the formal IND cover letter following `trial-ind/inputs/ReGARD-Regulatory-Guidance-for-Academic-Research-of-Drugs-and-Devices.md` (cover letter expectations) and the ReGARD instruction that the Cover Letter and FDA Form 1571 precede the Table of Contents. State the sponsor-investigator (Kevin Kawchak, ChemicalQDevice), the drug (daraxonrasib, RMC-6236), the serial number (0000, initial submission), the combined IND / IDE nature, the requested 30-day review, and the contents enclosed. Name the contact and reference the IND v1.0 and repository v4.3.0.

[DRAFTING INSTRUCTION] Render `trial-ind/mermaid/fig-01-ind-build-pipeline.md` and `fig-02-ind-toc-architecture.md` in the full stage as the orientation to how the submission is organized.

1 FDA Forms 1571 and 3674

1.1 FDA Form 1571 - Investigational New Drug Application

[DRAFTING INSTRUCTION] Summarize FDA Form 1571 from *trial-ind/inputs/FDA-1571instructions_{R1403-21-2023}.md* : its dual purpose (sponsor – investigator agreement and cover sheet for every submission); the contact / mailing fields; the I

1.2 FDA Form 3674 - Certification of Compliance

[DRAFTING INSTRUCTION] Summarize FDA Form 3674 from *trial-ind/inputs/ReGARD_{IND}Template.docx* : Box A (no clinical investigation), Box B (an investigation to which 42 U.S.C. §282(j), section 402(j) of the PHS Act, does not apply, the typical funded human trials).

[DRAFTING INSTRUCTION] Render *trial-ind/mermaid/fig-03-fda-1571-3674-flow.md* in the full stage.

2 Table of Contents

Cover Letter	1
1 FDA Forms 1571 and 3674	2
1.1 FDA Form 1571 - Investigational New Drug Application	2
1.2 FDA Form 3674 - Certification of Compliance	2
2 Table of Contents	3
3 Introduction	5
3.1 Introductory Statement	5
3.1.1 Name of the Drug and All Active Ingredients	5
3.1.2 Pharmacological Class of the Drug	5
3.1.3 Structural Formula of the Drug	5
3.1.4 Formulation of the Dosage Form(s) to be Used	5
3.1.5 Route of Administration	5
3.1.6 Duration of the Proposed Clinical Investigation(s)	5
3.2 Summary of Previous Human Experience	5
3.3 Status of Drug in Other Countries	5
3.4 Overview of Preclinical Data	5
3.5 References	6
4 General Investigational Plan	7
4.1 Rationale	7
4.2 Indication to be Studied	7
4.3 General Approach for Evaluation of Treatment	7
4.4 Description of First Year Trial(s)	7
4.5 Number of Subjects to be Evaluated	7
4.6 Drug Related Risks	7
4.7 References	7
5 Investigator Brochure	8
6 Proposed clinical research	9
6.1 Study Protocol	9
6.2 Informed Consent	9
6.3 Investigator and Facilities Data	9
6.3.1 FDA Form 1572	9
6.3.2 Disclosure of Financial Interests	9
7 Chemistry, Manufacturing and Control Information	10
7.1 Chemistry, Manufacturing and Control	10
7.1.1 Drug Substance	10
7.1.2 Drug Product	10
7.1.3 Placebo Product	10
7.1.4 Labeling	10
7.2 Environmental Assessment	10
8 Pharmacology and Toxicology Information	11
8.1 Pharmacology and Drug Distribution	11
8.1.1 Pharmacology Summary and Conclusions	11
8.1.2 Toxicology: Integrated Summary	11
8.1.3 Toxicology: Full Data Tabulation	11

9 Previous Human Experience	12
9.1 Marketed experience	12
9.2 Prior Clinical Research Experience	12
9.3 Clinical Care Experience	12
9.4 References	12
10 Additional Information	13
10.1 Drug Dependence and Abuse Potential	13
10.2 Radioactive Drugs	13
10.3 Pediatric Studies	13
10.4 Other Information	13
10.5 Selected References	13
11 Relevant Information	14
References and Back Matter	15
Acknowledgments	15
Author and ORCID	15
Ethical Disclosures	15
Data Availability	15
Rights and Permissions	15
Cite This Article	15
References	16

3 Introduction

3.1 Introductory Statement

[DRAFTING INSTRUCTION] Open with the investigational plan overview from `trial-protocol/final-protocol/publication/sections/sec-01-summary.tex` and `sec-02-introduction.tex`: a Phase 1, first-in-human, combined IND / IDE study of on-premises LLM-directed eight-arm robotic Whipple with perioperative daraxonrasib in KRAS G12-mutated PDAC. State the unmet need (5-year OS under 13 percent) and the acceleration thesis from `trial-documents/final-paper/publication`. Render `fig-06-document-landscape-before-during-after.md`.

3.1.1 Name of the Drug and All Active Ingredients

[DRAFTING INSTRUCTION] Daraxonrasib (RMC-6236), oral RAS(ON) multi-selective (pan-RAS) inhibitor. Source: `trial-protocol/.../sec-06-intervention.tex`. Render `fig-14-daraxonrasib-mechanism.md`.

3.1.2 Pharmacological Class of the Drug

[DRAFTING INSTRUCTION] RAS(ON) multi-selective small-molecule inhibitor binding GTP-bound mutant RAS via tri-complex with cyclophilin A. Source: protocol intervention section; `references.bib` (`daraxwhipple`, `onprewhipl`).

3.1.3 Structural Formula of the Drug

[DRAFTING INSTRUCTION] Describe daraxonrasib structurally (RMC-6236); state that the full structural formula is provided in the CMC section (§7.1.1).

3.1.4 Formulation of the Dosage Form(s) to be Used

[DRAFTING INSTRUCTION] Oral tablet, strengths for DL1 160, DL2 220, DL3 300 mg; child-resistant, light-protective container per 21 CFR §312.6. Source: CMC section, protocol.

3.1.5 Route of Administration

[DRAFTING INSTRUCTION] Oral, once daily, with a perioperative pause-and-restart advisory. Render `fig-18-perioperative-advisory.md`. Source: protocol intervention.

3.1.6 Duration of the Proposed Clinical Investigation(s)

[DRAFTING INSTRUCTION] Up to 28-day screening, surgery day 0, 30-day primary safety window, 90-day pathology, follow-up every 12 weeks to the 24-month OS endpoint. Source: `trial-protocol/.../sec-08-assessments.tex`.

3.2 Summary of Previous Human Experience

[DRAFTING INSTRUCTION] Daraxonrasib clinical program (RASolute 302, RASolve 301, Breakthrough Therapy June 2025) and the author's computational works. Render `fig-22-previous-human-experience.md` and `fig-12-daraxonrasib-document-thread.md`. Source: protocol introduction; `references.bib`.

3.3 Status of Drug in Other Countries

[DRAFTING INSTRUCTION] State marketing status and cross-jurisdictional alignment (EU MDR, NMPA, PMDA, MHRA, Health Canada, TGA) from `trial-protocol/.../sec-11-additional.tex`.

3.4 Overview of Preclinical Data

[DRAFTING INSTRUCTION] Summarize the pharmacology / toxicology basis and the digital-twin / QSP simulation evidence; full detail in §8. Render `fig-17-pharm-tox-structure.md`.

3.5 References

[DRAFTING INSTRUCTION] Cite the introduction sources via `\cite` keys from `references.bib`; the assembled list renders in §References and Back Matter.

4 General Investigational Plan

4.1 Rationale

[DRAFTING INSTRUCTION] State the scientific and schedule rationale: KRAS PDAC unmet need, the combined drug + device approach, and the acceleration argument (the six targets, the three timeline buckets, the patient cascade) from `trial-documents/final-paper/publication/sections/sec-04-results.tex` and `sec-05-discussion.tex`. Render `fig-04-six-acceleration-targets.md`, `fig-08-three-timeline-buckets.md`, and `fig-10-patient-time-saved-cascade.md`.

4.2 Indication to be Studied

[DRAFTING INSTRUCTION] Resectable or borderline-resectable KRAS G12-mutated PDAC; ECOG 0-1; Whipple candidate. Source: `trial-protocol/.../sec-05-population.tex`.

4.3 General Approach for Evaluation of Treatment

[DRAFTING INSTRUCTION] Phase 1, open-label, single-arm, 3+3 dose-finding, early-feasibility. Render `fig-15-three-plus-three-escalation.md` and `fig-21-objectives-endpoints.md`. Source: `protocol design/objectives`.

4.4 Description of First Year Trial(s)

[DRAFTING INSTRUCTION] Up to 18 treated across DL1-DL3, 1-2/month, single academic site, staggered sentinel enrollment. Source: `trial-protocol/.../sec-04-design.tex`.

4.5 Number of Subjects to be Evaluated

[DRAFTING INSTRUCTION] Up to n=18 treated (36 screened); design maximum 3 levels x 6. Source: `trial-protocol/.../sec-09-statistics.tex` (operating characteristics, Clopper-Pearson precision).

4.6 Drug Related Risks

[DRAFTING INSTRUCTION] Summarize daraxonrasib and device-related risks and mitigations, and the favorable benefit-risk balance for this low-survival population. Source: `trial-protocol/.../sec-02-introduction.tex` (risk-benefit framework); render `fig-10-patient-time-saved-cascade.md`.

4.7 References

[DRAFTING INSTRUCTION] Cite the plan sources via `references.bib`.

5 Investigator Brochure

[DRAFTING INSTRUCTION] Provide the Investigator's Brochure summary as a supporting document (use a clearly labeled supporting-document subsection). Summarize: the physical, chemical, and pharmaceutical properties of daraxonrasib; the nonclinical pharmacology, pharmacokinetics, and toxicology; the summary of data and guidance for the investigator; and the device description for the eight-arm robotic system. Sources: `trial-protocol/final-protocol/publication/sections/sec-06-intervention.tex` (drug and device), `sec-08-assessments.tex` (safety), and the CMC and pharmacology / toxicology sections of this IND (§7, §8). Render `fig-14-daraxonrasib-mechanism.md` and `fig-17-pharm-tox-structure.md`.

[DRAFTING INSTRUCTION] Note that the full standalone Investigator's Brochure is a separate bound document referenced here; this section provides the IND-level summary and a pointer to it (ReGARD: use separate sections to indicate and include supporting documents).

6 Proposed clinical research

6.1 Study Protocol

[DRAFTING INSTRUCTION] Summarize and include the Phase 1 study protocol as a supporting document: title, objectives (three co-primary, secondary, exploratory), design (3+3, $n \leq 18$), population, intervention (drug + eight-arm device), the perioperative advisory, assessments, statistics, and oversight. Sources: `trial-protocol/final-protocol/publication/sections/sec-01..sec-11`. Render `fig-21-objectives-endpoints.md`, `fig-18-perioperative-advisory.md`, and `fig-19-safety-reporting-triggers.md`. State that the full protocol (DOI 10.5281/zenodo.20780121) is the bound supporting document.

6.2 Informed Consent

[DRAFTING INSTRUCTION] Summarize the informed consent under 21 CFR part 50 and ICH E6(R3), including the Physical AI opt-out (21 CFR §312.60(f)) and the therapeutic- misconception handling. Source: `trial-protocol/.../sec-10-oversight.tex`.

6.3 Investigator and Facilities Data

[DRAFTING INSTRUCTION] Describe the single academic site, the multidisciplinary pancreatic program, and the device facilities. Render `fig-20-governance-oversight.md`.

6.3.1 FDA Form 1572

[DRAFTING INSTRUCTION] Summarize the Statement of Investigator (Form 1572): investigator qualifications, facilities, sub-investigators, and IRB. Note the executed form is assembled with the submission.

6.3.2 Disclosure of Financial Interests

[DRAFTING INSTRUCTION] State the sponsor-investigator financial-interest disclosure under 21 CFR part 54, the conflict of interest (CEO of the device manufacturer), and its mitigation by independent DSMB, ISM, and the Physical AI Safety Review Committee. Source: `trial-protocol/.../sec-10-oversight.tex`.

7 Chemistry, Manufacturing and Control Information

7.1 Chemistry, Manufacturing and Control

[DRAFTING INSTRUCTION] Introduce the CMC package and render `trial-ind/mermaid/fig-16-cmc-architecture.md`. Map to 21 CFR §312.23(a)(7).

7.1.1 Drug Substance

[DRAFTING INSTRUCTION] Daraxonrasib (RMC-6236): description, characterization, manufacturer, specification, and impurity controls. Source: `trial-protocol/.../sec-06-intervention.tex`.

7.1.2 Drug Product

[DRAFTING INSTRUCTION] Oral tablet at 160 / 220 / 300 mg: composition, manufacture, container-closure (child-resistant, light-protective, 21 CFR §312.6), and the stability program supporting shelf life.

7.1.3 Placebo Product

[DRAFTING INSTRUCTION] State not applicable: this is an open-label, single-arm study with no placebo. Per ReGARD, do not delete the section; note the non-applicability.

7.1.4 Labeling

[DRAFTING INSTRUCTION] Investigational caution statement (“Caution: New Drug - Limited by Federal law to investigational use”) and label content per 21 CFR §312.6.

7.2 Environmental Assessment

[DRAFTING INSTRUCTION] Claim a categorical exclusion under 21 CFR §25.31, or provide an environmental assessment if not excluded.

8 Pharmacology and Toxicology Information

8.1 Pharmacology and Drug Distribution

[DRAFTING INSTRUCTION] Introduce the pharmacology / toxicology package and render `trial-ind/mermaid/fig-17-pharm-tox-structure.md`. Map to 21 CFR §312.23(a)(8).

8.1.1 Pharmacology Summary and Conclusions

[DRAFTING INSTRUCTION] Summarize the RAS(ON) mechanism, pharmacodynamics, and drug distribution; the perioperative pharmacokinetic basis (32-iteration sweep, 0.45 ng/mL baseline trough). Source: `trial-protocol/.../sec-06-intervention.tex`; `references.bib` (`qspmetpancre`, `fdadigtwinpc`).

8.1.2 Toxicology: Integrated Summary

[DRAFTING INSTRUCTION] State the safety margins supporting the conservative DL1 160 mg first-in-human starting dose and the dose-limiting toxicity framework; reference the computational toxicology / QSP evidence.

8.1.3 Toxicology: Full Data Tabulation

[DRAFTING INSTRUCTION] Provide the full data tabulation as a supporting document; summarize the tabulated studies (digital-twin and QSP simulations, the 100,000-patient in silico Phase 3). Source: `references.bib` (`chatgpt100kp`, `pdacdigtwinp`).

9 Previous Human Experience

9.1 Marketed experience

[DRAFTING INSTRUCTION] State daraxonrasib's investigational status (not marketed); summarize the clinical program and Breakthrough Therapy designation (June 2025). Render `trial-ind/mermaid/fig-22-previous-human-experience.md`.

9.2 Prior Clinical Research Experience

[DRAFTING INSTRUCTION] Summarize RASolute 302 (Phase 3, pretreated metastatic PDAC) and RASolve 301 (first-line), and the broad pan-KRAS eligibility anchored to RASolute 302. Source: `trial-protocol/.../sec-02-introduction.tex`.

9.3 Clinical Care Experience

[DRAFTING INSTRUCTION] Summarize the surgical and oncologic care context (robotic Whipple, Dutch 2025 benchmark cohort) and the safety-reporting framework. Render `fig-19-safety-reporting-triggers.md` and `fig-13-post-submission-maintenance.md`. Source: `trial-protocol/.../sec-08-assessments.tex`.

9.4 References

[DRAFTING INSTRUCTION] Cite via `references.bib`.

10 Additional Information

10.1 Drug Dependence and Abuse Potential

[DRAFTING INSTRUCTION] State that daraxonrasib has no known dependence or abuse potential (not a controlled substance); brief justification.

10.2 Radioactive Drugs

[DRAFTING INSTRUCTION] State not applicable; daraxonrasib is not a radioactive drug.

10.3 Pediatric Studies

[DRAFTING INSTRUCTION] State the population is adults (age ≥ 18); summarize the pediatric plan / deferral rationale for this first-in-human oncology study.

10.4 Other Information

[DRAFTING INSTRUCTION] Summarize the Physical AI governance, cybersecurity, record retention (seven Physical AI record types, 21 CFR part 11), and the three-tier oversight. Render `trial-ind/mermaid/fig-20-governance-oversight.md`. Source: `trial-protocol/.../sec-10-oversight.tex`, `sec-11-additional.tex`.

10.5 Selected References

[DRAFTING INSTRUCTION] Cite via `references.bib`.

11 Relevant Information

[DRAFTING INSTRUCTION] Provide any additional relevant information requested by FDA and the method-level grounding and verification for this AI-generated IND: the five name-matching and grounding verifications and the figure-grounding discipline. Render `trial-ind/mermaid/fig-09-five-grounding-verifications.md` and `fig-11-figure-grounding-mermaid-to-tikz.md`. Source: `trial-documents/final-paper/publication/sections/sec-`

[DRAFTING INSTRUCTION] State the real-time auto-commit / auto-PR monitorability and the repository / directory / file name-matching (Rule 5), and that all LaTeX sources compile in Overleaf with pdfLaTeX.

References and Back Matter

[DRAFTING INSTRUCTION] Adapt the back matter from *trial-documents/final-paper/publication/sections/sec-08-Acknowledgments, Author and ORCID, Ethical Disclosures, Data Availability, Rights and Permissions (CC BY 4.0), and Cite This Article, then the bibliography. Keep black links; ensure every reference renders a clickable URL and a clickable DOI URL with DOI text, and that no link runs off the right margin.*

Acknowledgments

[DRAFTING INSTRUCTION] Credit Claude Code (Opus 4.8 Ultracode, 1M context) as the build model and thank ChatGPT 5.5 and Gemini 3.1 Pro; state IND v1.0, repository v4.3.0.

Author and ORCID

Kevin Kawchak, Chief Executive Officer, ChemicalQDevice. ORCID iD: <https://orcid.org/0009-0007-5457-8667>. Correspondence: kevink@chemicalqdevice.com.

Ethical Disclosures

[DRAFTING INSTRUCTION] State the independent-work disclaimer (no real patients, no PHI, no IRB approval sought; not an active IND; not medical or regulatory advice; not endorsed by FDA, NIH, HHS, IRB, ICH, or any sponsor).

Data Availability

[DRAFTING INSTRUCTION] State GitHub + Zenodo availability under CC BY 4.0, DOI 10.5281/zenodo.xxxxxxx.

Rights and Permissions

[DRAFTING INSTRUCTION] CC BY 4.0 statement with the license URL.

Cite This Article

[DRAFTING INSTRUCTION] Kawchak K. Phase 1 PDAC IND: AI Generation. Zenodo. 2026; 10.5281/zenodo.xxxxxxx.

References

- [1] K. Kawchak, "Phase 1 Pancreatic Cancer Trial: Efficient LLM Document Guidance." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 21018646. URL: <https://doi.org/10.5281/zenodo.21018646>.
- [2] K. Kawchak, "Oncology Trial PI LLM Adoption Guide." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20843290. URL: <https://doi.org/10.5281/zenodo.20843290>.
- [3] K. Kawchak, "A Phase 2, Daraxonrasib + LLM Guided Robotic PDAC Whipple Procedure, Clinical Trial Protocol." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20807027. URL: <https://doi.org/10.5281/zenodo.20807027>.
- [4] K. Kawchak, "On-Premises LLM-Directed Robotic Pancreaticoduodenectomy with Perioperative Daraxonrasib (RMC-6236) in KRAS-Mutated Pancreatic Ductal Adenocarcinoma." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20780121. URL: <https://doi.org/10.5281/zenodo.20780121>.
- [5] K. Kawchak, "Earning the Congress's Vote: A New Oncology Trial Framework for Enacting H. R. 9510." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20726461. URL: <https://doi.org/10.5281/zenodo.20726461>.
- [6] K. Kawchak, "Earning the Clinician's Trust: New Framework for Verified Physical AI Oncology Trials." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20710602. URL: <https://doi.org/10.5281/zenodo.20710602>.
- [7] K. Kawchak, "From H. R. 9510 to Federal Law: A Narrative Case for Verified Physical AI Oncology Trials." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20685379. URL: <https://doi.org/10.5281/zenodo.20685379>.
- [8] K. Kawchak, "H. R. 9510 (Bill v5.0) 2026." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20619762. URL: <https://doi.org/10.5281/zenodo.20619762>.
- [9] K. Kawchak, "H. R. 9510 (Bill v4.0) 2026." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20576907. URL: <https://doi.org/10.5281/zenodo.20576907>.
- [10] K. Kawchak, "H. R. 9510 (Bill v3.0) 2026." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20535429. URL: <https://doi.org/10.5281/zenodo.20535429>.
- [11] K. Kawchak, "Verification Before Generation Act of 2026." Zenodo, June 2026. DOI: 10. 5281/ zenodo. 20485580. URL: <https://doi.org/10.5281/zenodo.20485580>.
- [12] K. Kawchak, "VVUQ Physical AI Oncology Trial Bill." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20454870. URL: <https://doi.org/10.5281/zenodo.20454870>.
- [13] K. Kawchak, "Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20421754. URL: <https://doi.org/10.5281/zenodo.20421754>.
- [14] K. Kawchak, "VVUQ Oncology Clinical Trial LLM Verification Automation Priority over Existing Generated Code." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20372501. URL: <https://doi.org/10.5281/zenodo.20372501>.
- [15] K. Kawchak, "Threefold Humanoid 24/7 Adverse Event Oncology Trial Response Team: 4-Site Rotation." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20303281. URL: <https://doi.org/10.5281/zenodo.20303281>.
- [16] K. Kawchak, "2030: 60 Second Pancreatic Ductal Adenocarcinoma Robotic Whipple Procedure & Daraxonrasib Simulation." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20196639. URL: <https://doi.org/10.5281/zenodo.20196639>.
- [17] K. Kawchak, "2030: 60 Second Glioblastoma AI Robotic Surgery." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20113157. URL: <https://doi.org/10.5281/zenodo.20113157>.

- [18] K. Kawchak, "Patient Priority of Proposed U.S. Bills for Physical AI Oncology Clinical Trials." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 20045457. URL: <https://doi.org/10.5281/zenodo.20045457>.
- [19] K. Kawchak, "Accelerated Patient Prediction in Physical AI Oncology Clinical Trials: 4 Extensive Simulations." Zenodo, May 2026. DOI: 10. 5281/ zenodo. 19994945. URL: <https://doi.org/10.5281/zenodo.19994945>.
- [20] K. Kawchak, "Fully Automated Sponsor: Physical AI Oncology Clinical Trial Platform." Zenodo, Apr. 2026. DOI: 10. 5281/ zenodo. 19396256. URL: <https://doi.org/10.5281/zenodo.19396256>.
- [21] K. Kawchak, "National Platform for Physical AI Oncology Trials." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 19244918. URL: <https://doi.org/10.5281/zenodo.19244918>.
- [22] K. Kawchak, "Physical AI Oncology Clinical Trial Site Complete Documentation Package." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 19176370. URL: <https://doi.org/10.5281/zenodo.19176370>.
- [23] K. Kawchak, "A Cancer Patient's Journey Through a Regulated Physical AI Oncology Trial." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 19119939. URL: <https://doi.org/10.5281/zenodo.19119939>.
- [24] K. Kawchak, "Adaption: 21 CFR Part 312, End-to-End Physical AI Oncology Clinical Trial Unification." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 19057628. URL: <https://doi.org/10.5281/zenodo.19057628>.
- [25] K. Kawchak, "Adaption: 21 CFR Part 50, End-to-End Physical AI Oncology Clinical Trial Unification." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 19040707. URL: <https://doi.org/10.5281/zenodo.19040707>.
- [26] K. Kawchak, "Adaption: ICH Harmonised Guideline, End-to-End Physical AI Oncology Clinical Trial Unification." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 18973368. URL: <https://doi.org/10.5281/zenodo.18973368>.
- [27] K. Kawchak, "National MCP Servers for Physical AI Oncology Clinical Trial Systems." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 18916731. URL: <https://doi.org/10.5281/zenodo.18916731>.
- [28] K. Kawchak, "TrialMCP: MCP Servers for Physical AI Oncology Clinical Trial Systems." Zenodo, Mar. 2026. DOI: 10. 5281/ zenodo. 18870961. URL: <https://doi.org/10.5281/zenodo.18870961>.
- [29] K. Kawchak, "Federated Learning Physical AI Oncology Trials Unification." Zenodo, Feb. 2026. DOI: 10. 5281/ zenodo. 18795507. URL: <https://doi.org/10.5281/zenodo.18795507>.
- [30] K. Kawchak, "Unification Standard Level for Physical AI Oncology Trials." Zenodo, Feb. 2026. DOI: 10. 5281/ zenodo. 18778220. URL: <https://doi.org/10.5281/zenodo.18778220>.
- [31] K. Kawchak, "Code Generation Competition: 16 Proprietary vs. Open-Source LLMs & Iterative Learning Based on FDA Adverse Event Reporting System." Zenodo, Dec. 2025. DOI: 10. 5281/ zenodo. 18029100. URL: <https://doi.org/10.5281/zenodo.18029100>.
- [32] K. Kawchak, "AI Peer Review Acceleration of LLM-Generated Glioblastoma Clinical Trial Patient Matching ML, FDA/ICH/ISO, and FastAPI." Zenodo, Dec. 2025. DOI: 10. 5281/ zenodo. 17774560. URL: <https://doi.org/10.5281/zenodo.17774560>.
- [33] K. Kawchak, "LLM-Generated Glioblastoma Drug Synergy Machine Learning: From Rapid Code Prototypes to Project Deliverables Package." Zenodo, Nov. 2025. DOI: 10. 5281/ zenodo. 17614396. URL: <https://doi.org/10.5281/zenodo.17614396>.
- [34] K. Kawchak, "End-to-End Oncology Clinical Trial LLM Efficiency For Industry Adoption with FDA/ICH Regulations." Zenodo, Oct. 2025. DOI: 10. 5281/ zenodo. 17451709. URL: <https://doi.org/10.5281/zenodo.17451709>.
- [35] K. Kawchak, "Accelerating FDA Compliance and Cost Efficiency of in silico Clinical Trials via AI Digital Twin Pancreatic Cancer Simulation." Zenodo, Oct. 2025. DOI: 10. 5281/ zenodo. 17239510. URL: <https://doi.org/10.5281/zenodo.17239510>.

- [36] K. Kawchak, "QSP Metastatic Pancreatic Cancer AI Clinical Trial Simulation From Protocol to Prediction: Code, VVUQ, and Playbook." Zenodo, Aug. 2025. DOI: 10. 5281/ zenodo. 17001137. URL: <https://doi.org/10.5281/zenodo.17001137>.
- [37] K. Kawchak, "ChatGPT 100,000 Patient 24-Month In Silico Phase III 5-Arm Pancreatic Cancer Clinical Trial Triplicate." Zenodo, July 2025. DOI: 10. 5281/ zenodo. 16415815. URL: <https://doi.org/10.5281/zenodo.16415815>.
- [38] K. Kawchak, "End-to-End Pancreatic Ductal Adenocarcinoma Digital Twin Clinical Trial Proposals." Zenodo, June 2025. DOI: 10. 5281/ zenodo. 15735068. URL: <https://doi.org/10.5281/zenodo.15735068>.
- [39] K. Kawchak, "10 Year Glioblastoma Clinical Trial Meta-Analyses by Autonomous AI at Scale. Survival, HR, AE, and RoB scored in AI Reports and Charts, including Verifications." Zenodo, May 2025. DOI: 10. 5281/ zenodo. 15549830. URL: <https://doi.org/10.5281/zenodo.15549830>.
- [40] K. Kawchak, "AI revolution toward the cure of lung adenocarcinoma." Zenodo, Apr. 2025. DOI: 10. 5281/ zenodo. 15278152. URL: <https://doi.org/10.5281/zenodo.15278152>.
- [41] K. Kawchak, "Autonomous LLM Agent and scalable Reasoning LLM for generating cancer drug industry cost solutions." Zenodo, Mar. 2025. DOI: 10. 5281/ zenodo. 15072843. URL: <https://doi.org/10.5281/zenodo.15072843>.
- [42] K. Kawchak, "Cost containment of global monoclonal antibody drugs and cancer clinical trials via LLM focused reasoning." Zenodo, Feb. 2025. DOI: 10. 5281/ zenodo. 14968404. URL: <https://doi.org/10.5281/zenodo.14968404>.
- [43] K. Kawchak, "Clinical decision support based on Bevacizumab cancer trials and pushing the limitations of advanced LLMs." Zenodo, Jan. 2025. DOI: 10. 5281/ zenodo. 14968162. URL: <https://doi.org/10.5281/zenodo.14968162>.
- [44] K. Kawchak, "Cancer vs. Conversational Artificial Intelligence." bioRxiv, Dec. 2024. DOI: 10. 1101/2024. 12. 28. 630597. URL: <https://doi.org/10.1101/2024.12.28.630597>.
- [45] K. Kawchak, "mAb Bioprocess Engineering In-Context Table Forecasts using Conversational AI Literature Insight Generations." ChemRxiv, Dec. 2024. DOI: 10. 26434/ chemrxiv-2024- jzbj0. URL: <https://doi.org/10.26434/chemrxiv-2024-jzbj0>.
- [46] K. Kawchak, "Monoclonal Antibody Bioprocess Engineering Advancements Using Conversational Artificial Intelligence." ChemRxiv, Oct. 2024. DOI: 10. 26434/ chemrxiv-2024- 3m7m1. URL: <https://doi.org/10.26434/chemrxiv-2024-3m7m1>.
- [47] K. Kawchak, "Paclitaxel Biosynthesis AI Breakthrough." ChemRxiv, Oct. 2024. DOI: 10. 26434/ chemrxiv-2024- pqjd3. URL: <https://doi.org/10.26434/chemrxiv-2024-pqjd3>.
- [48] K. Kawchak, "High Dimensional and Complex Spectrometric Data Analysis of an Organic Compound using Large Multimodal Models and Chained Outputs." ChemRxiv, Sept. 2024. DOI: 10. 26434/ chemrxiv-2024- 06gf1. URL: <https://doi.org/10.26434/chemrxiv-2024-06gf1>.
- [49] K. Kawchak, "LMM Spectrometric Determination of an Organic Compound." ChemRxiv, Aug. 2024. DOI: 10. 26434/ chemrxiv-2024- qtnkj. URL: <https://doi.org/10.26434/chemrxiv-2024-qtnkj>.
- [50] K. Kawchak, "LMM Chemical Research with Document Retrieval." ChemRxiv, Aug. 2024. DOI: 10. 26434/ chemrxiv-2024- p91gm. URL: <https://doi.org/10.26434/chemrxiv-2024-p91gm>.